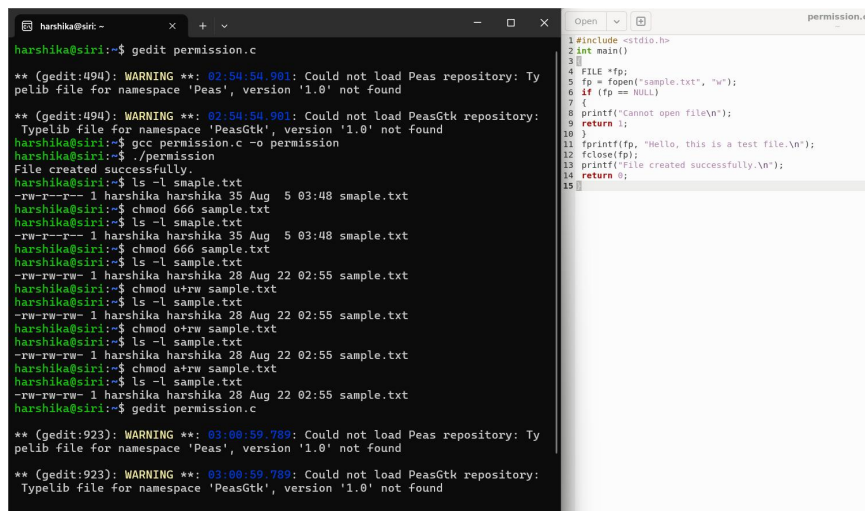


File Permissions



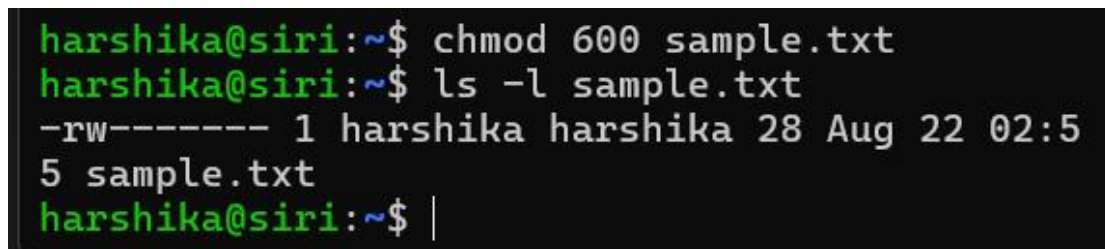
The image shows a terminal window on the left and a code editor on the right. The terminal window shows the user 'harshika@siri' creating a file 'permission.c' using 'gedit'. It then lists the file 'sample.txt' and sets permissions '666' using 'chmod'. The code editor shows the contents of 'permission.c', which is a simple C program that opens 'sample.txt' in write mode, prints a message, and closes the file.

```
harshika@siri:~$ gedit permission.c
** (gedit:494): WARNING **: 02:54:50.901: Could not load Peas repository: Ty
pelib file for namespace 'Peas', version '1.0' not found
** (gedit:494): WARNING **: 02:54:50.901: Could not load PeasGtk repository:
TypeLib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~$ gcc permission.c -o permission
harshika@siri:~$ ./permission
File created successfully.
harshika@siri:~$ ls -l sample.txt
-rw-r--r-- 1 harshika harshika 35 Aug 5 03:48 sample.txt
harshika@siri:~$ chmod 666 sample.txt
harshika@siri:~$ ls -l sample.txt
-rw-r--r-- 1 harshika harshika 35 Aug 5 03:48 sample.txt
harshika@siri:~$ chmod 666 sample.txt
harshika@siri:~$ ls -l sample.txt
-rw-rw-rw- 1 harshika harshika 28 Aug 22 02:55 sample.txt
harshika@siri:~$ chmod u+rw sample.txt
harshika@siri:~$ ls -l sample.txt
-rw-rw-rw- 1 harshika harshika 28 Aug 22 02:55 sample.txt
harshika@siri:~$ chmod o+rw sample.txt
harshika@siri:~$ ls -l sample.txt
-rw-rw-rw- 1 harshika harshika 28 Aug 22 02:55 sample.txt
harshika@siri:~$ chmod a+rw sample.txt
harshika@siri:~$ ls -l sample.txt
-rw-rw-rw- 1 harshika harshika 28 Aug 22 02:55 sample.txt
harshika@siri:~$ gedit permission.c
** (gedit:923): WARNING **: 03:00:59.789: Could not load Peas repository: Ty
pelib file for namespace 'Peas', version '1.0' not found
** (gedit:923): WARNING **: 03:00:59.789: Could not load PeasGtk repository:
TypeLib file for namespace 'PeasGtk', version '1.0' not found
```

```
1 #include <stdio.h>
2 int main()
3 {
4     FILE *fp;
5     fp = fopen("sample.txt", "w");
6     if (fp == NULL)
7     {
8         printf("Cannot open file\n");
9         return 1;
10    }
11    fprintf(fp, "Hello, this is a test file.\n");
12    fclose(fp);
13    printf("File created successfully.\n");
14    return 0;
15 }
```

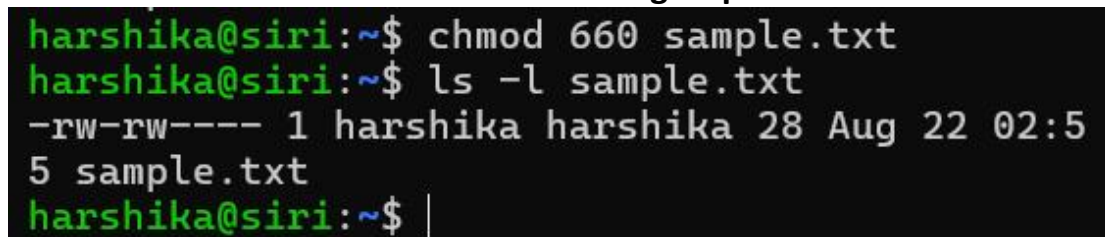
TASKS

1. Give Read and Write only to the owner



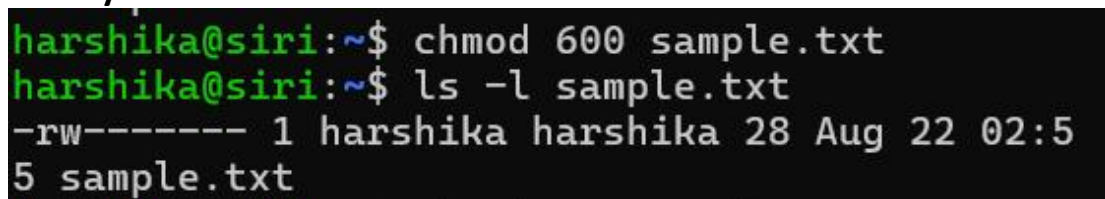
```
harshika@siri:~$ chmod 600 sample.txt
harshika@siri:~$ ls -l sample.txt
-rw----- 1 harshika harshika 28 Aug 22 02:5
5 sample.txt
harshika@siri:~$ |
```

2. Give Read and Write to the owner and group



```
harshika@siri:~$ chmod 660 sample.txt
harshika@siri:~$ ls -l sample.txt
-rw-rw---- 1 harshika harshika 28 Aug 22 02:5
5 sample.txt
harshika@siri:~$ |
```

3. Only owner can read and write



```
harshika@siri:~$ chmod 600 sample.txt
harshika@siri:~$ ls -l sample.txt
-rw----- 1 harshika harshika 28 Aug 22 02:5
5 sample.txt
```

4. Owner read/write, everyone else read-only

```
harshika@siri:~$ chmod 644 sample.txt
harshika@siri:~$ ls -l sample.txt
-rw-r--r-- 1 harshika harshika 28 Aug 22 02:5
5 sample.txt
```

5. Owner can do everything, others can only read and execute

```
harshika@siri:~$ chmod 755 sample.txt
harshika@siri:~$ ls -l sample.txt
-rwxr-xr-x 1 harshika harshika 28 Aug 22 02:5
5 sample.txt
```

6. Owner and group can read/write

```
harshika@siri:~$ chmod 660 sample.txt
harshika@siri:~$ ls -l sample.txt
-rw-rw---- 1 harshika harshika 28 Aug 22 02:5
5 sample.txt
```

7. Only owner can execute

```
harshika@siri:~$ chmod 700 sample.txt
harshika@siri:~$ ls -l sample.txt
-rwx----- 1 harshika harshika 28 Aug 22 02:5
5 sample.txt
harshika@siri:~$ |
```

FILL IN THE BLANKS

8. Owner: read + write; Group: nothing; Others: nothing → **600**

9. Owner: read + write; Group: read; Others: read → **644**

10. Owner: read + write + execute; Group: read + execute; Others: read + execute → **755**

11. Owner: read + write; Group: read + write; Others: nothing → **660**

12. Everyone: read only → **444**